

List of Figure Captions & Symbols (Nomenclature)

Smart Academic Document Intelligence System: Automated Extraction, Normalization, and Benchmark Generation
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1. List of Figure Captions (Separate Sheet for Typesetting)

Fig. 1. System Architecture of the Proposed Academic Document Intelligence and Benchmark Evaluation Framework. The architecture decomposes into two decoupled subsystems: (1) Academic Document Benchmark Generator (ADBG v1.0) for vector template compilation and multi-profile optical degradation, and (2) AU DIC Evaluation Subsystem for read-only neural inference ingestion, semantic canonical normalization, and nine-class OCR error taxonomy classification.

Fig. 2. Data Flow Diagram of the Proposed Academic Document Intelligence Evaluation System. Traces the end-to-end multi-level transformation lifecycle across Level 0 (Context Level), Level 1 (Framework Execution Flow), and Level 2 (Diagnostic Error Classification & Evaluation Engine).

Fig. 3. Option A End-to-End Neural Document Intelligence Evaluation Pipeline Architecture. Demonstrates direct image pixel tensor ingestion via Ollama local runtime to MiniCPM-V (7.6B Q4_0) without external OCR pre-segmentation.

Fig. 4. Accuracy Improvement after Semantic Canonical Normalization. Bar chart illustrating field-level accuracy and exact match gains across Certificate, Marksheet, and ID Card document categories before and after normalization.

Fig. 5. Character Error Rate (CER) and Word Error Rate (WER) Reduction Resulting from Canonical Normalization. Visualizes the 90.42% relative reduction in Character Error Rate (from 38.13% to 3.65%) and 90.53% reduction in Word Error Rate (from 285.31% to 27.01%).

Fig. 6. Total False-Negative Field Mismatches Resolved by Each Individual Domain Normalizer Rule. Horizontal distribution of the 2,620 corrected errors across Date, Roll Number, Degree Alias, Numeric, Honorific/Whitespace, and University Alias normalizers.

Fig. 7. Field-by-Field Accuracy Improvement Comparing Raw String Matching Against Canonical Normalization. Detailed multi-field breakdown comparing baseline unnormalized exact match rates against post-canonicalization accuracy.

Fig. 8. Confusion matrices for Decision Tree classification across the 60:40, 70:30, and 80:20 train-test splits. Evaluates the predictability of document extraction failure modes using axis-aligned decision trees (93.69% accuracy, 95.91% F1, 0.8303 MCC).

Fig. 9. Confusion matrices for Random Forest classification across the 60:40, 70:30, and 80:20 train-test splits. Compares ensemble bagging classification across 24,480 observations against decision tree models.

2. List of Mathematical Symbols & Notations (Nomenclature)

Symbol	Mathematical / Technical Meaning	Unit / Domain
P	Field-level entity Precision ($P = TP / (TP + FP)$)	Ratio [0, 1]
R	Field-level entity Recall ($R = TP / (TP + FN)$)	Ratio [0, 1]
F1	Harmonic mean of Precision and Recall	Percentage [0, 100%]
CER	Character Error Rate $((S + D + I) /$	Percentage [0, 100%]

	N_chars via Levenshtein distance)	
WER	Word Error Rate ((Sw + Dw + Iw) / N_words)	Percentage [0, inf)
Raw EM	Raw Exact Match rate (unnormalized string equality)	Percentage [0, 100%]
Norm EM	Normalized Exact Match rate (post-canonicalization string equality)	Percentage [0, 100%]
Joint EM	Document-level Joint Record Exact Match	Percentage [0, 100%]
N(.)	Six-stage semantic canonicalization operator	Transformation Function
I(.)	Indicator function (1 if condition holds, 0 otherwise)	Binary {0, 1}
TP, FP, FN	True Positive, False Positive, and False Negative counts	Integer ≥ 0
S, D, I	Substitutions, Deletions, and Insertions in character edit distance	Integer ≥ 0
chi ²	McNemar's chi-squared test statistic for paired binary proportions	Statistical Statistic
W	Wilcoxon signed-rank test statistic for paired non-parametric differences	Statistical Statistic
t	Paired Student's t-test statistic	Statistical Statistic
p	Statistical significance probability (p-value)	Probability [0, 1]
alpha	Hypothesis rejection significance threshold (alpha = 0.01 or 0.05)	Significance Level
B	Number of non-parametric bootstrap resampling iterations (B = 10,000)	Integer Count
N	Total evaluated field observations (N = 24,480)	Integer Count
D	Total evaluated document specimens (D = 360)	Integer Count